ID	Feature	User / Technical Story	Acceptance Criteria	Story Points
Sprint 1 (22 Sep to 20 Oct): Data, model and app				**83**
US_01	Data Pipeline	As the development team, we need clean per-eye labels, a patient-level split and preprocessed photos, so the model can be trained and tested fairly.	Per-eye labels taken from each eye's own keywords pass the verification script; patients are split 70/15/15 with seed 42; one shared preprocessing function runs on every photo; code is merged into main.	23
US_02	Dual-Head Model	As a screening technician, I want one photo to give both a cataract and a glaucoma result, so I can triage both conditions in one step.	A shared ResNet with two heads trains end to end on the real labels, weights rare cases, never reads external data, and passes the smoke test.	16
US_03	External Test Sets	As a reviewer, I want the model tested on outside datasets it never trained on, so I can trust that it works beyond ODIR-5K.	Test-only labels for ORIGA, DRISHTI-GS, ACRIMA and retina_dataset_2016 have checked counts; the overlap check passes; the metrics protocol is written; calibration code is ready and uses the validation split only.	15
US_04	App Prototype (carried over to Sprint 2)	As a screening technician, I want to upload a patient's fundus photo and see a clear cataract and glaucoma result, so I can decide quickly whether the patient should see an eye specialist.	Upload, Results and History screens work; upload shows a preview and rejects non-images; results show separate cataract and glaucoma results from the API, with "Uncertain, refer" as a normal result; mock data only; follows the DUAL branding.	18
US_05	Photo Quality Check	As a screening technician, I want the app to stop photos that can't be screened and tell me why, so I don't get a result from an unusable file.	Unreadable, almost-black and non-fundus uploads are blocked with a clear message; hazy or blurry eye photos still go to the model; all ODIR training photos pass; 17 automated tests pass.	3
US_06	Model Ablation	As the development team, we want cataract-only, glaucoma-only and shared two-head models trained with identical settings, so we can test whether sharing the backbone helps.	Nine models are saved (3 model types x 3 seeds); the training split is used to learn and the validation split to stop; settings for every run are recorded; no external data is used.	8
Sprint 2 (planned)				**28**
US_07	Calibration & Abstention	As an optometrist, I want the tool to say "low image quality or uncertain, refer" when it is not sure, so that I do not act on a weak result.	Calibration, one decision threshold per disease and one abstention cutoff are fitted on the validation split only; the frozen values are committed and not changed afterwards.	5
US_08	Internal Test	As the development team, we want a within-dataset baseline, so that external results can be compared against it.	All three model types are scored once on the internal test split after calibration is frozen; AUC with 95% confidence interval for each disease; results table is in the repository.	3
US_09	External Testing	As a primary care physician, I want to know how the model performs on photos from other clinics, so that I can judge how far to trust its referrals.	ORIGA and DRISHTI-GS are scored with the frozen thresholds and calibration; ACRIMA and retina_dataset_2016 are reported in their own rows; AUC with confidence intervals and calibration for each set.	5

ID	Feature	User / Technical Story	Acceptance Criteria	Story Points
US_10	Technical Paper Draft	As a reader of the paper, I want the datasets and limitations described with exact counts, so that I can judge the results.	Dataset and limitations sections use counts from the verification script; the cataract external-testing gap is stated as a limitation; PDF and Word versions are on the wiki.	5
US_11	Cataract Haze Check	As a primary care physician, I want to know the cataract result reflects the lens and not a blurry photo, so that I can trust a cataract referral.	The outcome that means "the model detects haze" is written down before results are seen; cataract share is reported for media-opacity and normal eyes; blur curve is plotted against real cataract eyes.	5
US_16	Testing and Review	As the team, we want every Sprint 2 story tested and reviewed against its acceptance criteria by someone who did not build it, so that we know each story really works before we close it.	Test cases are written for each Sprint 2 story; each is run and recorded as pass or fail with the date; every pull request for a Sprint 2 story is reviewed before it is merged; problems found are logged in Jira and linked to the story.	5
Sprint 3 (planned)				**29**
US_12	Glaucoma Heatmap Check	As a primary care physician, I want to see that a glaucoma result is based on the optic disc, so that I can trust the referral.	Grad-CAM heatmaps are produced for ORIGA and DRISHTI-GS; the share of images whose heatmap peak falls inside the disc outline is reported per dataset.	5
US_13	Ablation Comparison	As a reader of the paper, I want to know whether sharing one backbone helps, hurts or makes no difference, so that the design choice is justified.	Single-disease and shared models are compared per disease on the internal test and, for glaucoma, on the external tests; spread across seeds is shown; table and conclusion are in the repository.	3
US_14	Real Model in App	As an optometrist, I want to upload a fundus photo and get real cataract and glaucoma risk scores, so that I can decide whether to refer.	The app runs on the trained model; "Uncertain, refer" is shown as a normal result; every results table names the commit that produced it; the results commit is tagged before the final demo.	8
US_15	Final Documentation	As a developer joining later, I want install, deployment and API manuals, so that I can run and extend the app.	Data card, user manual, deployment manual and API documentation in PDF and Word; final technical paper with its history page; reviewed by the data pipeline and external validation roles.	8
US_17	Testing and Review	As the team, we want every Sprint 3 story tested and reviewed against its acceptance criteria by someone who did not build it, so that the final product we demo and hand over really works.	Test cases are written for each Sprint 3 story; each is run and recorded as pass or fail with the date; the full app is tested end to end with the real model before the final demo; every pull request is reviewed before it is merged; problems found are logged in Jira.	5
			Total	**140**